Aaron Lin

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EDUCATION

Boston University Boston, MA
B.S. Mechanical Engineering, Aerospace Concentration Sep 2024 – May 2028
Materials Science & Engineering minor | GPA: 3.76/4.0, Dean's List
Study Abroad: BU Sydney Sophomore Engineering Program, University of Sydney Spring 2026
Relevant Coursework: Mechanics of Materials, Energy and Thermodynamics, Differential Equations, Computational Linear Algebra

PROFESSIONAL EXPERIENCE

Mechanical Engineering Intern Feb 2026 – Jun 2026
Culturon (plasma-activated coatings for life sciences) Sydney, Australia
- Designed and executed Optical Emission Spectroscopy (OES) experiments using an HR4000 spectrometer; authored protocols formalizing plasma consistency evaluation and elemental contaminant detection procedures.
- Developed an automated Python analysis suite (pandas, scipy, astroquery) to process OES spectral data, isolate elemental emission peaks, and cross-reference identifications against the NIST atomic line database.
- Authored a benchmarking report comparing OES plasma diagnostics against an established reference chamber, establishing key emission line signatures for contamination tracking and process qualification.

AI Trainer Oct 2025 – Dec 2025
Handshake AI Fellowship Remote
- Developed and evaluated domain-specific prompts benchmarking LLM accuracy across engineering and physical science subfields; applied subject-matter expertise to identify reasoning failures on scientific and quantitative topics.
- Provided structured expert feedback on model outputs, supporting targeted accuracy improvements on technical and domain-specific knowledge tasks.

PROJECTS

Project Engineer Sep 2024 – Present
Boston University Rocket Propulsion Group Boston, MA
- Designing an aerodynamic fairing for an externally-mounted downcomer on the Argo rocket; evaluating tangent ogive and power-law nosecone profiles under Mach 2.81 max-Q conditions via ANSYS Fluent CFD and OpenRocket trajectory simulation.
- Designed a converging-diverging rocket nozzle in SolidWorks using 1D compressible flow theory; modeled pressure loading via FEA under 80 psi chamber pressure, predicting a minimum factor of safety of 2.12 in the pressure-only model for design review.
- Fabricated the nozzle via resin 3D printing and executed a static test-fire, integrating with a 1-inch NPT chamber and relay ignition circuitry to demonstrate integrated operation under live combustion conditions.

Academic Project Sep 2025 – Dec 2025
Deskinator — Autonomous Cleaning Robot Boston, MA
- Authored Arduino C++ firmware implementing S-curve pathing, IR proximity-triggered 90° edge-avoidance turns, and gesture-activated start to enable fully autonomous operation across a 4 ft by 4 ft desk surface.
- Contributed to full system assembly and iterative subsystem redesigns, including upgrading from stepper to 12V N20 encoder motors and replacing the vacuum fan with a 5300 RPM, 230 CFM axial unit after prototype testing revealed insufficient torque and suction.
- Team ranked in top 4 of approximately 20 groups; achieved 76% debris pickup across suction trials with the upgraded fan and nozzle configuration.

Academic Project Jan 2025 – May 2025
Temperature Sensor Prototype Boston, MA
- Developed and programmed an Arduino-based room temperature monitor in C++, featuring an LCD display, TMP36 sensor, and an alert system (LEDs, buzzer) housed in a custom ABS enclosure.
- Authored a technical report evaluating system performance, identifying key limitations (battery life, component selection, manufacturing process), and proposing improvements for potential commercialization.

SKILLS

Software: SolidWorks (Simulation, HSMworks), Fusion, Onshape, ANSYS Fluent, OpenRocket, Python, MATLAB, C++, Java
Engineering: Mechanical Design, FEA, CFD, Supersonic Aerodynamics, 1D Compressible Flow, Stress Analysis, GD&T (fundamentals), Thermodynamic Cycle Analysis, Technical Writing, Systems Integration, Experimental Testing
Fabrication: Additive Manufacturing (FDM, Resin), CNC Machining, Manual Lathe, Manual Mill, Soldering, Electronics Assembly
Languages: English (Native), Mandarin Chinese (Conversational)

CERTIFICATIONS

A Hands-On Introduction to Engineering Simulations — edX / CornellX, 2026
Machine Learning Specialization — DeepLearning.AI / Stanford University, 2024